Analysis of Algorithms
Lab 08

1. Is it possible to exchange numeric values of two variables, say, u and v, without using any extra storage? Show how with a program.
2. Implement the *ComparisonCountingSort* algorithm and test it.
3. Implement the *DistributionCountingSort* algorithm and test it. Compare it with the *ComparisonCountingSort* from previous point. What are the differences of complexity and implementation?
4. Implement a hashing algorithm using C++ (no dictionaries or libraries allowed). Use closed hashing as the collision resolution method.

Submission

Upload your solutions as a PDF file. In case of coding assignments, upload to GitHub and include the link in the PDF file. Your submission should include screenshots of the execution of your code when applicable.